Traffic Congestion and the Price of Anarchy: A Data-Driven Analysis of London's Road Network

Networks I - Concepts and Algorithms

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Abstract

We examine traffic inefficiencies in London's road network through a data-driven approach to calculate the Price of Anarchy (PoA). London has consistently ranked among the world's most congested cities. We use a comprehensive empirical data on traffic from OpenStreetMap (OSM) and the UK Department for Transport (DfT) to develop and apply a unique PoA formula that captures the relationship between observed travel times and theoretical optimal routing. Our methodology employs spatial nearest-neighbor matching to associate traffic monitoring points with corresponding road segments, enabling precise congestion analysis across London's diverse road network. Our methodology involves mapping traffic data to the road network and applying spatial analysis techniques to identify congestion hotspots and inefficiency trends across different types of roads and in Boroughs. We complement our PoA analysis with Nash equilibrium simulations on a representative network sample, confirming substantial potential time savings through optimal routing strategies.

1 Introduction

Modern urban transportation systems face one big challenge: traffic congestion. This phenomenon leads to increased travel times, fuel consumption, and pollution. London has been consecutively ranked as one of the slowest cities in the world. TomTom (2023) reported that a 10 km journey in central London takes on average 37 minutes and 20 seconds, which places London in first place as the slowest city in the world. There have been a wide variety of policy interventions, such as congestion charges and Ultra Low Emission Zones (ULEZ); however, traffic inefficiencies still persist in major cities. Understanding and quantifying traffic congestion has been a topic of interest for research with the ultimate goal of improving urban mobility. One of the key metrics used to quantify these inefficiencies is the Price of Anarchy (PoA), which measures the impact of selfish routing compared to an optimal, system-wide traffic distribution.

We use datasets from OSM and the UK DfT in order to integrate real-time traffic flow measurements, vehicle counts, and road attributes to analyze congestion levels and compute PoA. Our approach focuses on empirical PoA estimation for the city of London, providing a data-driven perspective on how traffic inefficiencies vary across different road types and speed limits.

2 Literature Review

The Price of Anarchy (PoA) is a concept widely used in the literature to quantify inefficiencies in decentralized transportation networks. Several studies have proposed different formulas for the PoA with the common goal of quantifying inefficiencies in decentralized transportation networks. Although the core objective is the same — comparing the cost of selfish routing to an optimal, system-coordinated scenario — each study defines PoA differently.

Roughgarden and Tardos (2002) were among the first to analyze traffic congestion modeling and the PoA. Their model is based on Wardrop Equilibrium, where individual drivers minimize their own travel time without considering the overall efficiency of the

system. The authors used the following PoA specification:

$$\text{PoA} = \frac{\text{Total Cost}_{UE}}{\text{Total Cost}_{SO}}$$

The authors established an upper bound for the PoA in networks with affine cost functions, proving that the worst-case inefficiency is at most 4/3. Despite being useful, this model has limitations, as it assumes rational decision-making, which may not always align with real-world driver behavior. Nevertheless, this work serves as a foundation for game-theoretic traffic analysis and continues to influence modern research on transportation optimization and network efficiency.

Correa et al. (2008) introduced a geometric approach to analyze network inefficiencies in several real-world cities. The authors suggested the following formula for the PoA in order to demonstrate that the network structure plays a crucial role in determining traffic inefficiencies:

$$\text{PoA} = \frac{C_{eq}}{C_{opt}}$$

Where C_{eq} represents the total cost at equilibrium, reflecting congestion under selfish routing, and C_{opt} represents the socially optimal cost, where routes are assigned to minimize overall congestion. The authors proved that selfish routing increases total travel time by up to 30%, that network topology affects PoA, and that removing certain roads can surprisingly reduce congestion. London's infrastructure and traffic patterns have evolved significantly since 2008. Correa et al. (2008) study is limited by theoretical equilibrium models and does not incorporate real-time data or speed limits, these are aspects our study aims to include.

More recently, Kuang et al. (2021) extended the PoA framework by incorporating externalities. The main acknowledgment of this study is that congestion on one route can spill over and affect other interacting routes. The authors' approach captures network-wide congestion effects, which makes it particularly relevant for busy cities such as London. The authors define PoA as:

$$\text{PoA}(X) = \frac{(b_{t1} - b_{t2})^2}{4a_1a_2t_1t_2 - (b_{t1} + b_{t2})^2 + 1}$$

The numerator represents the difference in the demand for traffic between two interacting routes, capturing how uneven distributions can lead to inefficiencies. On the other hand, the denominator incorporates the combined effects of congestion sensitivity (a_1, a_2) and free-flow travel times (t_1, t_2), which ensures that externalities are considered as inefficiency factors. The additional constant term $+1$ accounts for baseline inefficiencies even in the absence of extreme externalities.

The PoA formula proposed by Kuang et al. (2021) shows that spillover effects between two routes can significantly worsen overall inefficiency (by increasing the PoA). The authors prove that traditional PoA model, which fails to consider externalities, may be underestimating the real-world congestion effects. Despite revolutionary, an important limitation of this study lies with the difficulty in estimating externality parameters, where real-world traffic dynamics involves numerous unpredictable factors.

Cominetti et al. (2024) addressed an important limitation of classical PoA models that assume a single origin-destination (OD). The authors considered multi-commodity interactions in transportation networks. In real-world environments, multiple OD pairs

interact, and congestion in one part of the network can influence traffic flow elsewhere. Taking this into consideration the authors define PoA as:

$$\text{PoA}(\mu) = \frac{SC^*(\mu)}{SC_{SO}(\mu)}$$

Where $SC^*(\mu)$ represents the social cost at equilibrium under decentralized decision-making, and $SC_{SO}(\mu)$ represents the socially optimal cost, where routing decisions are centrally optimized. The authors quantify how PoA changes with demand shifts, assuming that inefficiencies are not constant but vary depending on traffic density, network structure, and demand elasticity. The study emphasizes the need for adaptive traffic policies that respond to changing demand conditions, which replaces the notion of a static upper bound for PoA. While this non-linear dynamic model successfully captures traffic patterns, it is computationally intensive, particularly for large-scale networks.

Table 1: Comparison of PoA Formulations with Our Approach

Aspect	**Wardrop Equilibrium-Based PoA (Roughgarden & Tardos, 2002)**	**Multi-Commodity Routing PoA (Cominetti et al., 2024)**	**Geometric PoA (Correa et al., 2008)**	**PoA with Externalities (Kuang et al., 2021)**	**Our PoA Approach**
Definition	$\text{PoA} = \frac{\text{Total_Cost}_{\text{UE}}}{\text{Total_Cost}_{\text{SO}}}$	$\text{PoA}(\mu) = \frac{\text{SC}^*(\mu)}{\text{SC}^{\text{SO}}(\mu)}$	$\text{PoA} = \frac{\text{C}_{\text{eq}}}{\text{C}_{\text{opt}}}$	$\text{PoA}(X) = 1 + \frac{(b_1t_1 - b_2t_2)^2}{4a_1t_1a_2t_2 - (b_1t_1 + b_2t_2)^2}$	$\text{PoA} = \frac{\text{Actual_Time}}{\text{Free-flow_Time}}$
Traffic Behavior	Drivers selfishly minimize personal travel time (Wardrop Equilibrium)	Models interactions across different OD pairs	Network topology-based geometric analysis	Accounts for spillover effects between routes	Uses observed real-world travel times
Optimal Benchmark	Min-cost flow optimization	Piecewise differentiable equilibrium functions	Compares equilibrium cost to optimal conditions	Models external congestion impact	No congestion (maximum speed limits)
Mathematical Basis	Potential function analysis in game theory	Nonlinear programming with demand-sensitive equilibrium	Graph-theoretic geometric representations	Externality models for congestion pricing	Empirical speed and traffic volume data
Key Findings	Upper bound PoA $\leq$ 4/3 for affine cost functions	PoA fluctuates with demand shifts	Network topology determines inefficiency	Higher PoA with external congestion effects	Captures real-world inefficiencies directly
Pros	Simple, widely accepted in theoretical research	Suited for analyzing fluctuating demand	Intuitive visual representation	Accounts for congestion spillovers	Directly applicable to urban planning
Cons	Assumes rational decision-making	Computationally intensive	Limited real-time data integration	Difficult parameter estimation	Doesn't model equilibrium optimizations

3 Data and Methodology

OpenStreetMap (OSM) is one of the primary datasets that is used to obtain the most up-to-date and accurate characteristics regarding the map and roads of London. The OSM data are extracted using the OSMnx Python library and are essential for linking traffic data from other sources to actual road segments using important data columns such as the longitude and latitude coordinates for the roads, their speed limits, and connectivity within the road system. The UK Department for Transport (DfT) website provides the other two primary datasets that are used to gather vehicle and traffic data, specifically for

the year 2023. The first is named annual average daily flow (AADF) traffic data, where the annualized average is collected using a combination of manual traffic counts, inductive loop sensors, and automatic traffic counters at certain locations throughout the city. The second is named raw traffic count data, which are collected using real-time periodic traffic monitoring from different sources such as automatic number plate recognition (ANPR) cameras, loop sensors, and GPS-based tracking systems. There are several columns from each data source, but some of the most crucial to the analysis are *latitude, longitude* for geographic coordinates of the traffic monitoring point, *link_length_km* for the length of the road segment, *all_motor_vehicles* for traffic volume counts, *maxspeed* for the speed limit on the specific road segment, and *geometry* for the geometric representation of the road in LineString format to use for spatial analysis and mapping.

We use real-world traffic counts with attributes of the road network in order to be able to compute congestion levels and inefficiencies. After preprocessing and cleaning the data we merge traffic count data to road segments. The merging process is done using a BallTree algorithm, which allows to use spatial nearest-neighbor matching in order to associate each traffic monitoring point to its closest road within the OSM data and correctly align the two. To compute congestion ratios and the price of anarchy on the merged datasets, we use the following equations:

Key Formulas used

1. **Observed Speed**:

$$\text{Observed Speed} = \frac{\text{Total Travel Time}}{\text{Total Distance}} = \frac{\text{link_length_km} \times 1000}{\text{all_motor_vehicles}}$$

2. **Free-flow Time**:

$$\text{Free-flow Time} = \frac{\text{Road Length (km)}}{\text{Speed Limit (km/h)}}$$

3. **Delay**:

$$\text{Delay} = \text{Observed Travel Time} - \text{Free-Flow Time}$$

4. **Congestion Ratio**:

$$\text{Congestion Ratio} = \frac{\text{Observed Travel Time}}{\text{Free-Flow Travel Time}} = \frac{\left(\frac{\text{link_length_km}}{\text{observed speed}}\right)}{\left(\frac{\text{link_length_km}}{\text{speed_limit}}\right)}$$

5. **Price of Anarchy (PoA)**:

$$\text{PoA} = \frac{\text{Total Free-Flow Travel Time}}{\text{Total Observed Travel Time}}$$

4 Analysis

Traffic Monitoring Points and Congestion Ratio

In this section, we present our results from the congestion ratios and price of anarchy computations. To initialize our analysis, we first look at a map of London and try to

better comprehend its road network and its distribution of traffic monitoring points. To do so, we use the road network data that is provided from the OSM source and are depicted as the gray lines, which is then overlaid with the spatial distribution of the traffic monitoring points provided from the AADF data and depicted as the red lines, as illustrated in Figure 1, a geospatial map depicting traffic monitoring points on top of London's road network. From this visual, several observations can be made such as there being a higher density of monitoring location points within central London due to the higher traffic volume of being a city center. As we move away from the center, traffic monitoring points become more dispersed, as suburban areas have fewer data to collect. In addition to the city center, major highways and roads are also seen to have clusters of monitoring location points, highlighting the need for traffic management. From this figure, we can confirm that the traffic data aligns with the city's major roadways and conclude that central and high-traffic areas require more coverage with the use of more monitoring points.

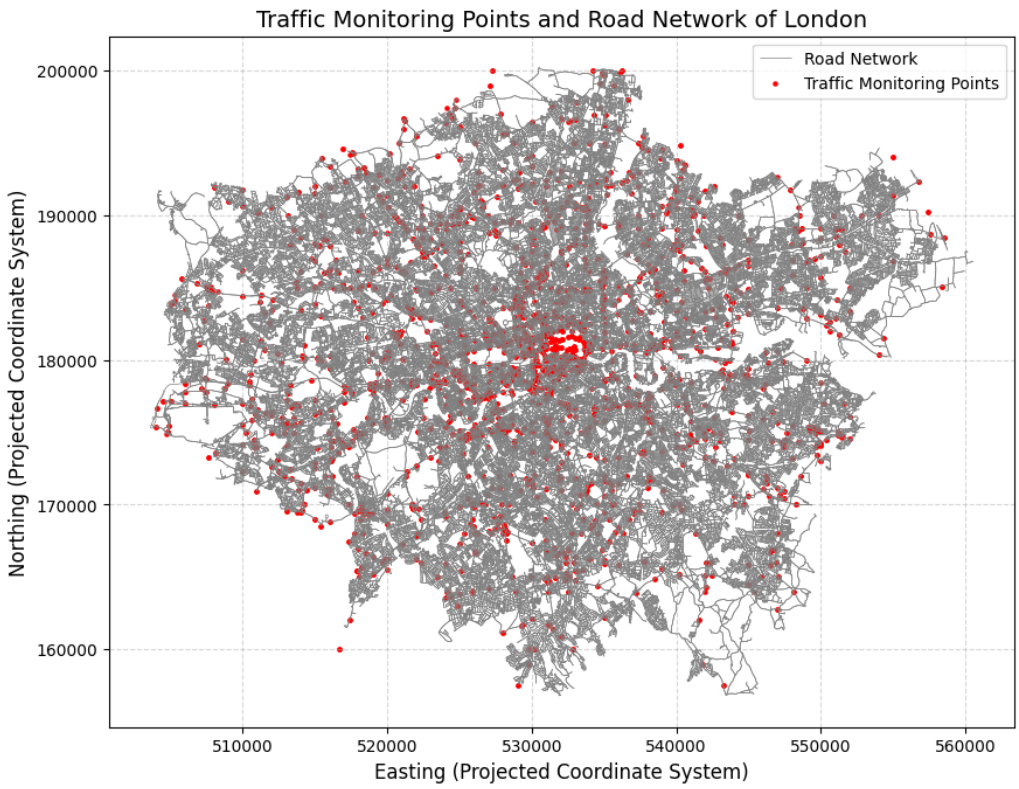


Figure 1: Traffic Monitoring Points and Road Network of London

To continue with the analysis, we now use the merged dataframe in order to compute the congestion ratio between different types of road and their speed limits. By computing the congestion ratio, it allows one to capture the real impact of travel times within realistic versus more ideal scenarios. In order to obtain a better comparison between different speed limits, a comparison was made where roads with a 20 mph speed limit present an average ratio of approximately 3,95, while roads with higher speed limits present a higher average ratio of approximately 10,17. That is to say, roads with speed limits greater than 20 mph endure more than double the delay and take roughly ten times as long as they would under uncongested conditions. Figure 2, a bar chart to visually compare congestion levels based on speed limits and counts, reinforces these observations by visually depicting the

distribution of congestion ratios for the two categories. Those roads with a 20 mph speed limit cluster around the ratio between 3 and 4 and those roads with higher speed limits cluster within a greater range between 10 and 12. This part of the analysis highlights how having higher speed limits does not inherently alleviate traffic issues and delays, but rather coincides with more severe congestion levels. These findings help us to conclude that speed alone does not guarantee efficiency within an urban network, but rather a combination of other factors as well.

These findings are also reinforced in Figure 3, a traffic congestion hotspot plot in London, where the congestion ratios are used to depict the congestion hotspots. The ratios are grouped by different severity levels, low, moderate, high, or severe. A clear pattern can be seen within the city center, where there is a dense cluster of orange and red points to depict how congestion is both more frequent and intense in this area. In comparison to the outer suburban areas, although there are still some red dots distributed throughout these areas, the ratio of severe congestion levels is small in comparison to the combined ratio of moderate and high congestion levels. The severe levels of congestion and traffic burden seen in central areas amplify inefficiencies under traffic conditions during the year 2023, and this can be due to different possible reasons such as a higher population density, complex road layouts, or commercial activity. With a persistent challenge such as this, targeted interventions are necessary where London has implemented key measures such as congestion charging zones, improving public transportation and cycle infrastructure, and converting more roads to have a 20 mph limit as that has already been linked to less congestion. Thus, higher speed limits do not inherently reduce travel time, but do the opposite and increase traffic congestion, reinforcing the need to improve or create new targeted measures to mitigate severe congestion in urban centers.

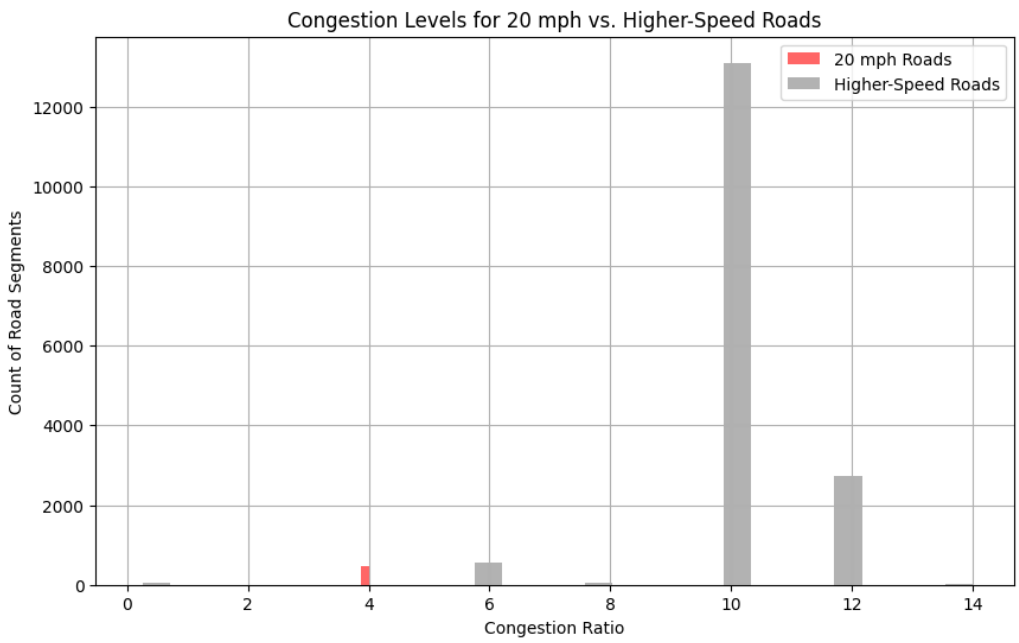


Figure 2: Congestion Levels for 20 mph vs. Higher-Speed Roads

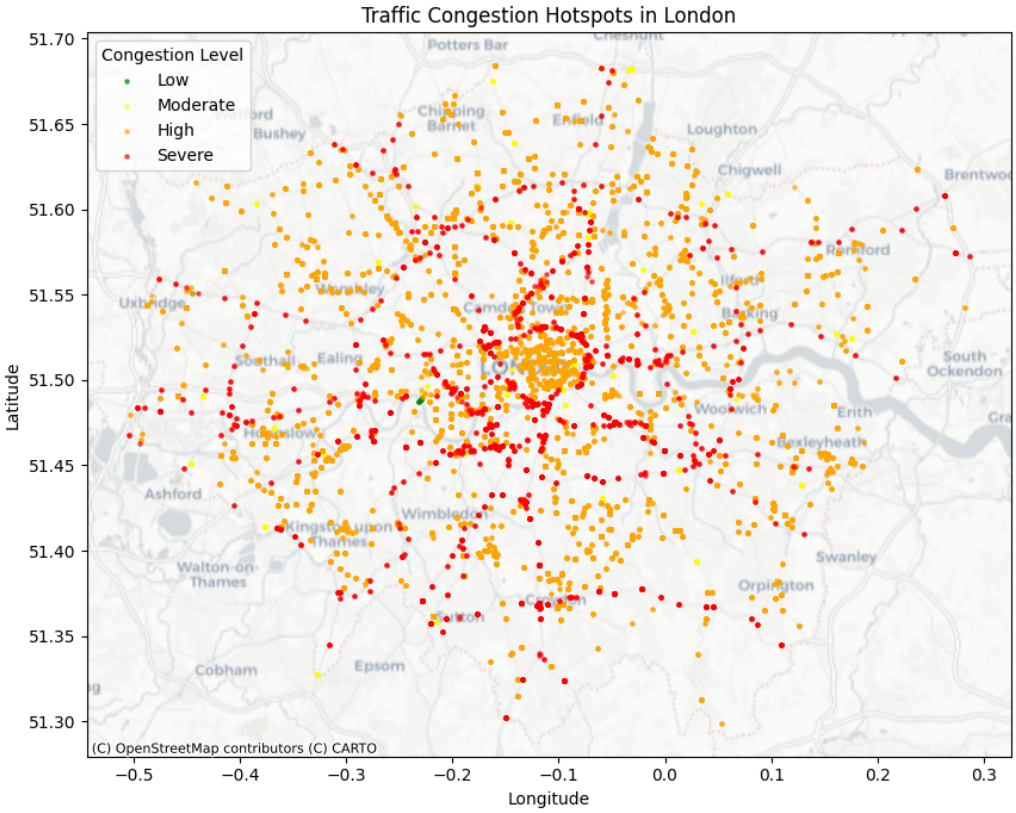


Figure 3: Traffic Congestion Hotspots

Price of Anarchy

Following our analysis of quantifying congestion levels and delays in different speed limits, we proceed to measure the overall inefficiency of the road network by calculating the price of anarchy. As already stated, this crucial metric helps to understand the behavior of individual drivers and their driving decisions that impact traffic inefficiencies. Keeping in mind the reality that drivers act selfishly most of the time, it is essential to calculate the inefficiency and analyze congestion hotspots. Although the congestion ratio measures how long a trip takes compared to an ideal scenario, the price of anarchy, on the other hand, extends this concept to system-wide inefficiencies. Figure 4, which visualizes the relationship between the congestion ratio and the price of anarchy, shows a strong positive correlation between the two, where an increase in congestion will also cause an increase in PoA and inefficiency. In an ideal scenario, the PoA would equal 1, however, this is not the case here because roads in real-world scenarios experience significant deviations from this ideal value.

A spatial representation of PoA across London's road network can be seen in Figure 5. As expected, the roads surrounding the city center present road segments with some of the highest PoA values, as they are seen with a darker shade of red. This indicates that actual travel times exceed those travel times under optimal conditions. Within the actual city center, some points that have low levels of PoA colored blue can be seen, possibly because when people are able to reach the center they are then able to park and move around the city more easily by walking or public transportation. PoA values are seen to become smaller in value as well as more dispersed the further away from the center it is. Some areas outside the city center show high PoA values, which can indicate

localized inefficiencies that may need to be addressed. Overall, these findings align with the congestion trends we found and demonstrate that higher PoA values coincide with areas of more frequent congestion and those with longer delays.

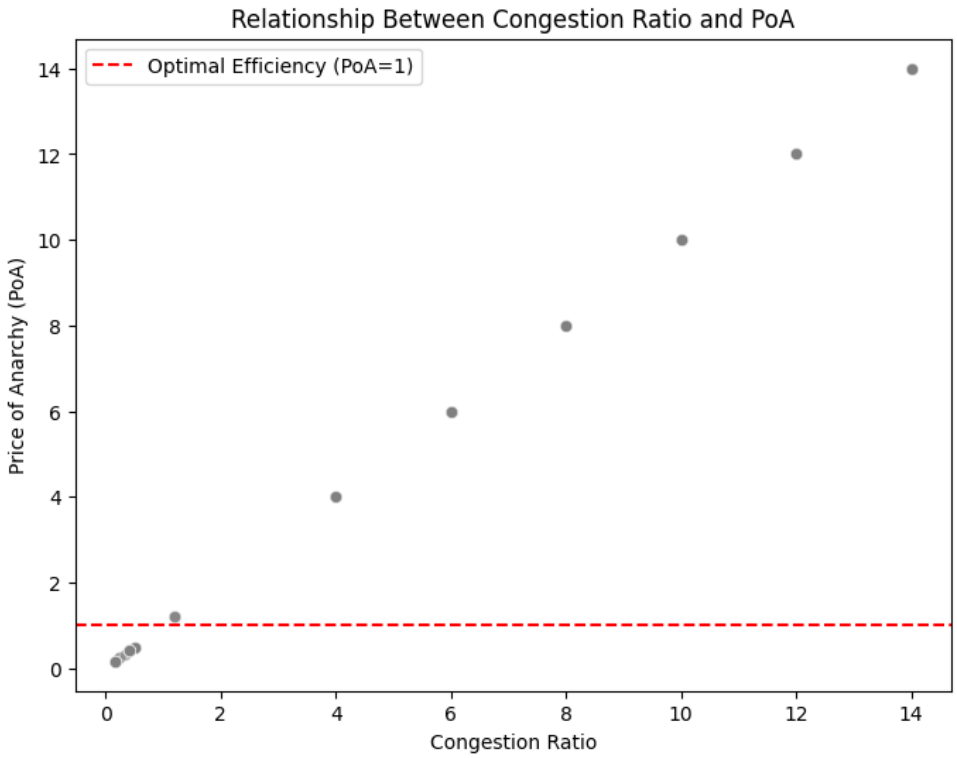


Figure 4: Relationship Between Congestion Ratio and PoA

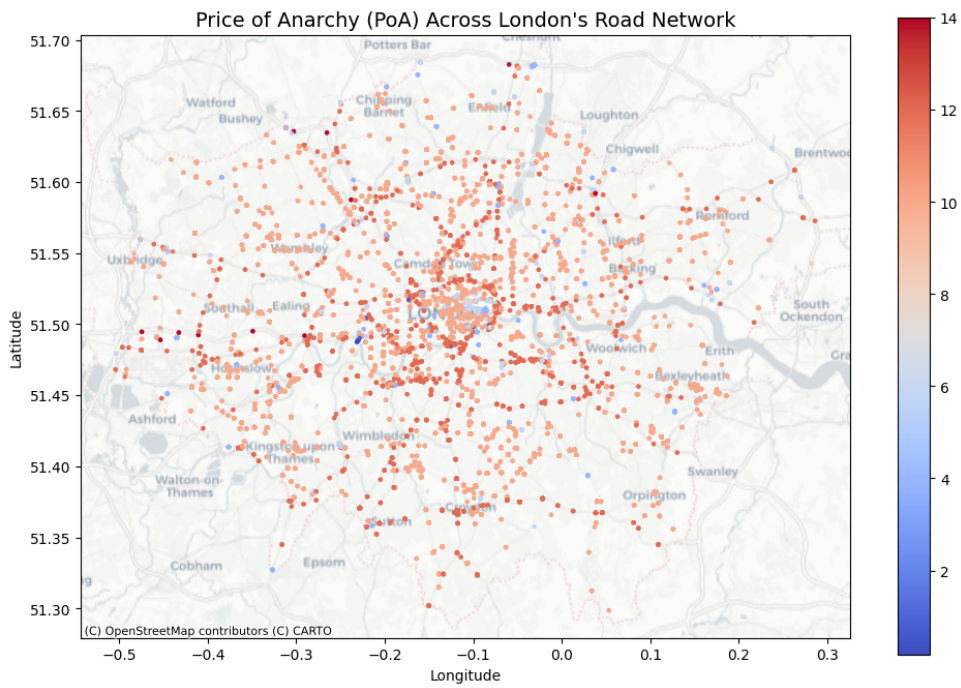


Figure 5: Price of Anarchy (PoA) Across London's Road Network

In order to adjust PoA values based on congestion values to analyze traffic inefficiencies, the PoA column is multiplied by the congestion ratio column to get the weighted PoA. Figure 6 shows the distribution of the congestion-weighted price of anarchy across the road network where the first clear observation is that there are clear peaks around 100 and 150. This suggests that a certain group of road segments experience similar inefficiencies and confirms that this is a common type of traffic condition or a common road usage pattern and not just random. Those roads with low congestion-weighted PoA illustrate efficient traffic conditions despite there still being some congestion. In comparison to the standard PoA distribution, Figure 6 illustrates a skewed distribution to highlight persistent congestion patterns within road segments. This analysis also further supports the argument that traffic inefficiency seen in urban road networks is not just due to traffic flow choices, but is also based on congestion disproportionately influencing certain routes as a result.

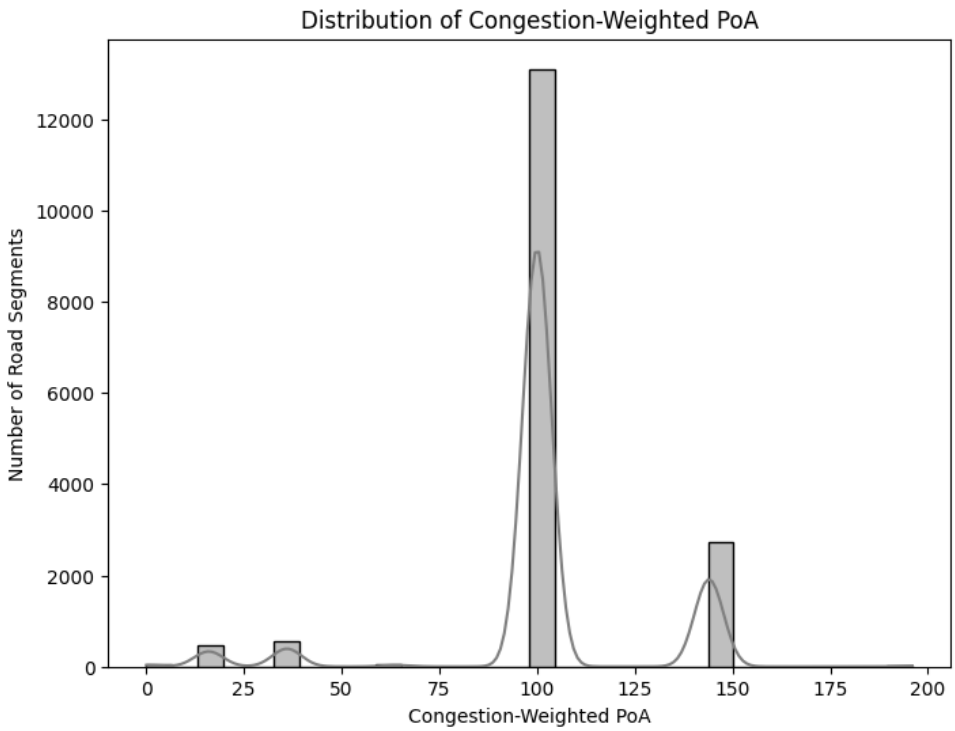


Figure 6: Distribution of Congestion-Weighted PoA

PoA Analysis by Borough's in London

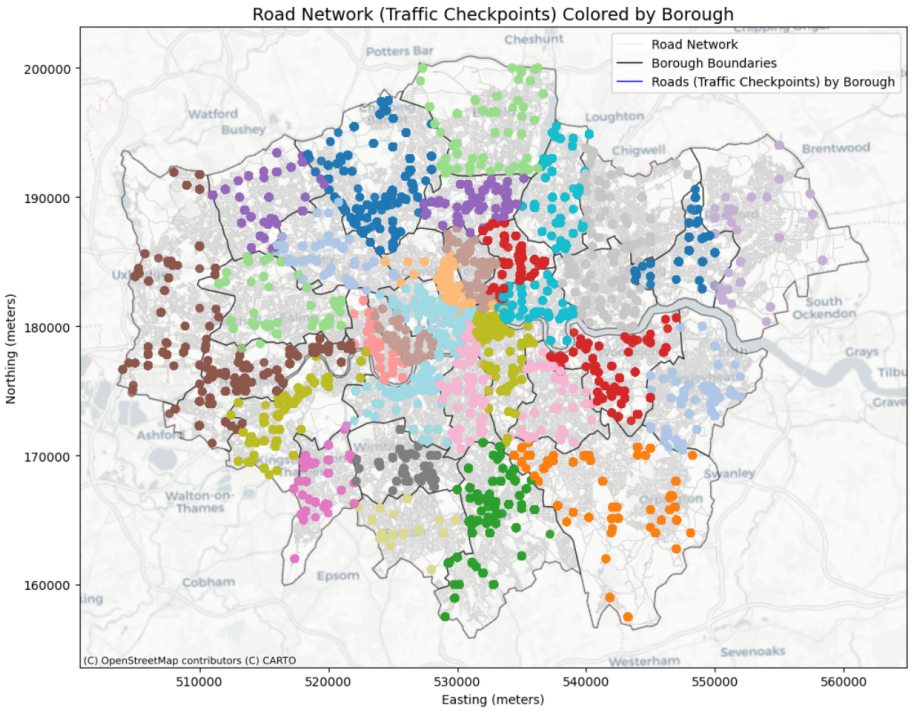


Figure 7: Traffic checkpoints and Roads across Boroughs

Overview of Traffic Checkpoints and Road Network by Borough
Understanding the distribution of traffic congestion requires examining the spatial distribution of monitoring points and road networks across London's boroughs. The image above in Figure 7 visually represents the road network (edges) and traffic monitoring checkpoints (nodes) colored by borough.

- **Borough Boundaries**: The boroughs are delineated with black lines, showing how traffic monitoring points are distributed.
- **Traffic Checkpoints**: The colored dots indicate the locations of traffic monitoring points where vehicle counts and speeds were measured.
- **Traffic Network Representation**: The gray road network shows the primary transportation arteries, with some areas having denser networks and a higher concentration of checkpoints.

Central boroughs (e.g., City of London, Westminster, Camden, Southwark) have a high density of checkpoints, reflecting their high traffic volume. In contrast, outer boroughs (e.g., Sutton, Wandsworth, Bromley) have sparser monitoring points, which may contribute to differences in estimated congestion levels.

Understanding the Price of Anarchy (PoA)
The Price of Anarchy (PoA) measures the inefficiency of decentralized traffic routing compared to an optimal system-wide traffic distribution. It is calculated as:

$$\text{PoA} = \frac{\text{Total Free-Flow Travel Time}}{\text{Total Observed Travel Time}}$$

Note - Optimal Efficiency (PoA = 1) occurs when the observed travel times match the free-flow travel times (i.e., no congestion). Higher PoA Values (¿1) indicate significant inefficiencies due to congestion, with drivers experiencing much longer travel times than in an optimal scenario.

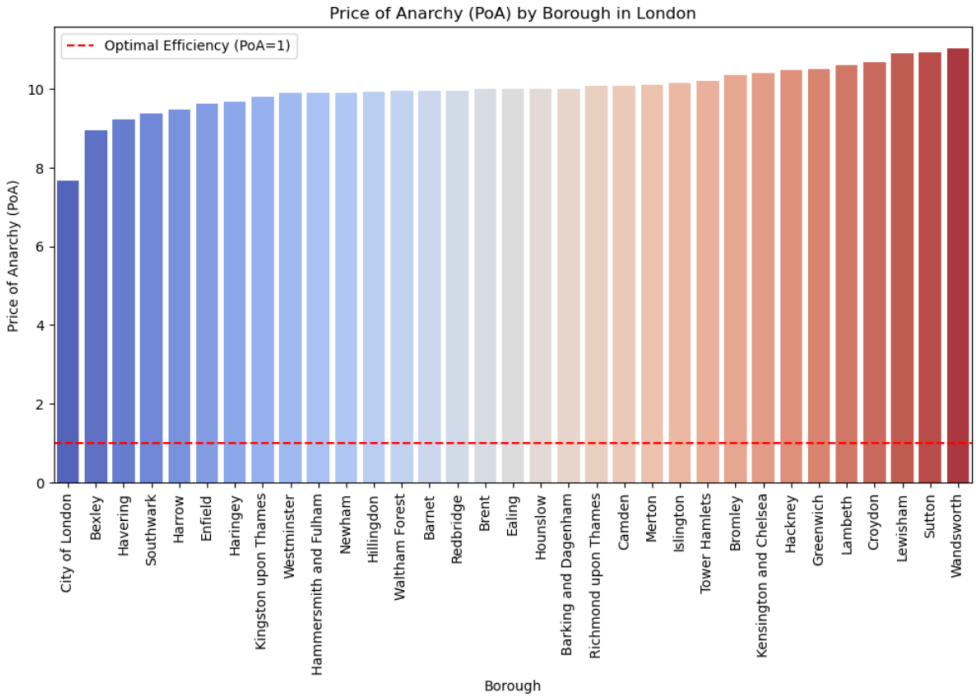


Figure 8: PoA by Boroughs

The bar chart provided illustrates the PoA for each London borough. The key observations are:

- **City of London has the lowest PoA (7.65)**: Possible Explanation: The result looks counterintuitive, but in recent years City of London has enforced high congestion charges, invested in extensive public transport, and a pedestrian-friendly environment, which can explain regulated traffic with improved relative efficiency. High employment density means many commuters use alternative transport options such as the London Underground.

- **Highest PoA Values (Above 10) in Wandsworth, Sutton, and Lewisham**: These boroughs have longer delays relative to free-flow conditions, indicating severe congestion. Wandsworth (PoA = 11.03) and Sutton (PoA = 10.93): These outer boroughs likely experience significant commuter traffic, with limited high-capacity road infrastructure leading to delays. Lewisham (PoA = 10.90): A dense residential area with limited road expansion options and reliance on arterial roads, which can become bottlenecks.

The red dashed line in the chart represents PoA = 1, indicating the ideal scenario with no traffic inefficiencies. All boroughs significantly exceed this threshold, confirming that selfish routing behaviors contribute heavily to congestion inefficiencies.

Factors Contributing to PoA Variations Across Boroughs

- **Congestion Control Measures**: Boroughs with congestion charges and low-emission zones (e.g., City of London) tend to have lower PoA values.
- **Infrastructure and Public Transport Accessibility**: Areas with high public transport penetration experience lower traffic inefficiencies.
- **Road Network Density**: Sparse road networks in outer boroughs can lead to higher PoA, as alternative routes are limited.
- **Commuter Patterns**: Boroughs with heavy inbound commuter traffic (e.g., Wandsworth, Sutton) experience higher congestion, raising PoA values.

The Price of Anarchy analysis provides valuable insights into how traffic inefficiencies vary across London's boroughs. The findings highlight the need for targeted traffic policies, such as improved public transport connectivity and demand-responsive congestion measures, to mitigate inefficiencies in high-PoA boroughs.

By understanding PoA variations, policymakers can implement better congestion management strategies to optimize urban mobility and reduce travel inefficiencies in London.

Selfish, Optimal Routes and Nash Equilibriums

To understand the impact of selfish routing behaviors on London's road network, we conducted a detailed analysis of Nash equilibrium, system-optimal routing, and potential time savings. This section explores the dynamic interplay between driver behavior and traffic efficiency through computational network analysis.

Sampling and Network Construction

Our analysis focused on a representative subset of London's road network to make the computational analysis tractable while maintaining statistical validity. Given the computational complexity of routing algorithms, analyzing the entire London road network would be prohibitively expensive in terms of processing time and memory requirements. We employed a random sampling approach to select 50,000 nodes from the complete network, using a fixed random seed to ensure reproducibility. The sampled nodes were selected randomly from the union of all origin and destination points in the original network. This sampling approach reduced the edge count from 301,003 to 187,874, creating a subset of London's road network. The resulting directed graph contained 186,014 edges and 103,380 nodes.

To quantify the inefficiency of selfish routing decisions, we developed two distinct cost functions that capture different routing behaviors:

1. **Selfish Cost Function (User Equilibrium)**: This function models how individual drivers make routing decisions to minimize their own travel time without considering the impact on other drivers. We leveraged our previously calculated Price of Anarchy (PoA) metric as a direct multiplier of free-flow travel time. Since PoA represents the ratio of actual travel time to free-flow travel time, multiplying free-flow time by PoA yields the congested travel time experienced under selfish routing conditions. This approach elegantly incorporates real-world congestion effects while maintaining computational simplicity.

2. **Optimal Cost Function (System Optimum)**: This function represents an idealized scenario where all routes are assigned to minimize total system-wide travel time. Unlike the selfish cost function, it incorporates both the base travel time and a marginal cost component that accounts for the externalities a driver imposes on others when using a particular road segment. The marginal cost factor (set at 0.1 in our model) represents the rate at which congestion increases with additional flow relative to road capacity. This factor would ideally be calibrated using empirical traffic flow data, representing a potential area for refinement in future research.

The key distinction between these functions lies in their treatment of externalities. While the selfish cost function only considers the driver's own travel time, the optimal cost function incorporates a marginal cost component that accounts for the impact of a driver's route choice on other road users. This difference is fundamental to understanding the gap between Nash equilibrium (where all drivers use selfish routing) and system optimum (where routes are assigned to minimize total travel time).

Nash Equilibrium Analysis

We computed Nash equilibrium travel times by simulating traffic assignment based on selfish routing decisions. The simulation process began by identifying origin-destination (OD) pairs within our sampled network and assigning traffic volume based on the AADF (Annual Average Daily Flow) data. For each OD pair, we used Dijkstra's shortest path algorithm to determine the likely route a driver would take when making selfish routing decisions.

Once routes were determined, we updated the flow on each edge along the path proportionally to the traffic demand between the corresponding origin and destination. The travel cost for each edge was then recalculated using our selfish cost function, which incorporates the Price of Anarchy to reflect congestion effects.

Our simulation assigned traffic to only 975 origin-destination pairs within the selected high-traffic nodes, despite working with a sample of 5,000 nodes. This unexpectedly low number of OD pairs represents a significant limitation in our analysis that warrants further investigation. Several potential explanations exist:

1. **Network Connectivity Issues**: The random sampling process may have disconnected many potential origin-destination pairs, leaving only 975 that maintained valid paths in the sampled network.

2. **Data Filtering Effects**: The requirement that both origin and destination nodes be present in the high-traffic node set may have been too restrictive, particularly if traffic monitoring points were primarily positioned at origins or destinations but rarely at both.

3. **Node Attribution Problems**: There may be issues with how traffic monitoring points were mapped to network nodes, causing most OD pairs to be incorrectly excluded from analysis.

4. **Limited Matching Between Datasets**: The original AADF data might have contained relatively few complete OD pairs that could be successfully mapped to the OSM road network data.

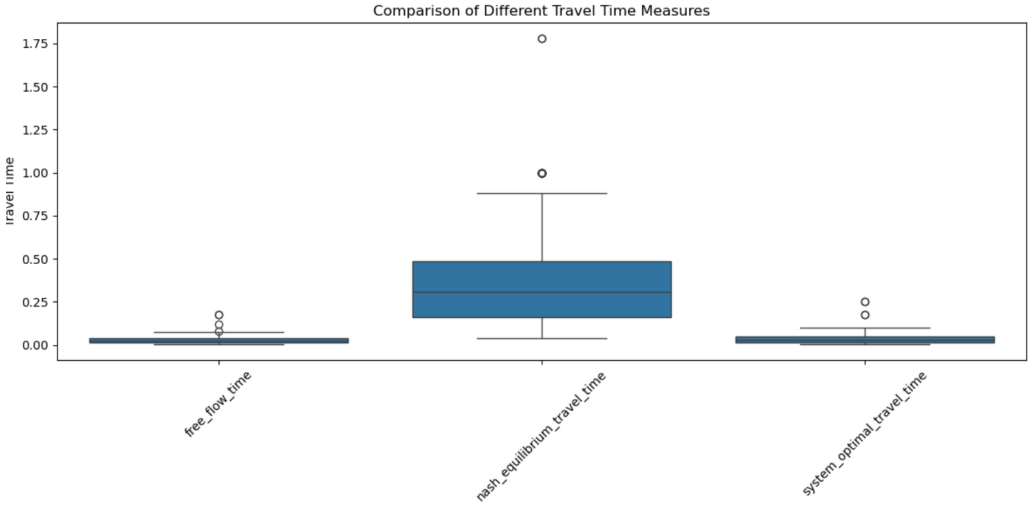
5 Key Findings and Limitations

Our analysis reveals several insights about London's traffic patterns. We categorized our findings into three main areas: current traffic conditions, Nash equilibrium metrics, and potential improvements through optimal routing.

Current Traffic Conditions

The current state of London's road network shows severe congestion levels. The average congestion ratio of 10.05 indicates that travel times in London are approximately 10 times longer than they would be under free-flow conditions.

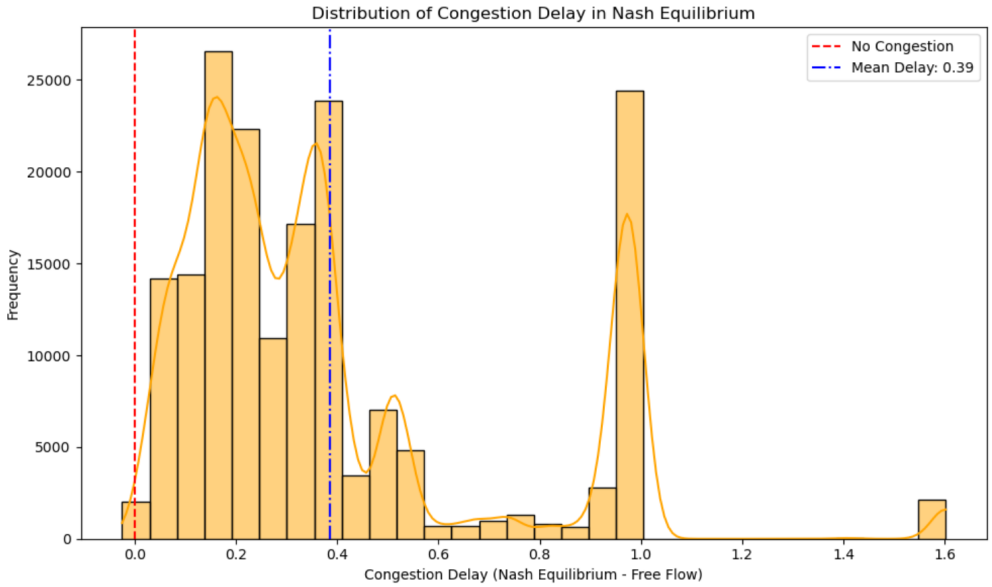
The average congestion delay per route segment of 0.28 units might seem modest, but when accumulated across the entire journey and multiplied by thousands of vehicles, it represents a significant economic and environmental cost.

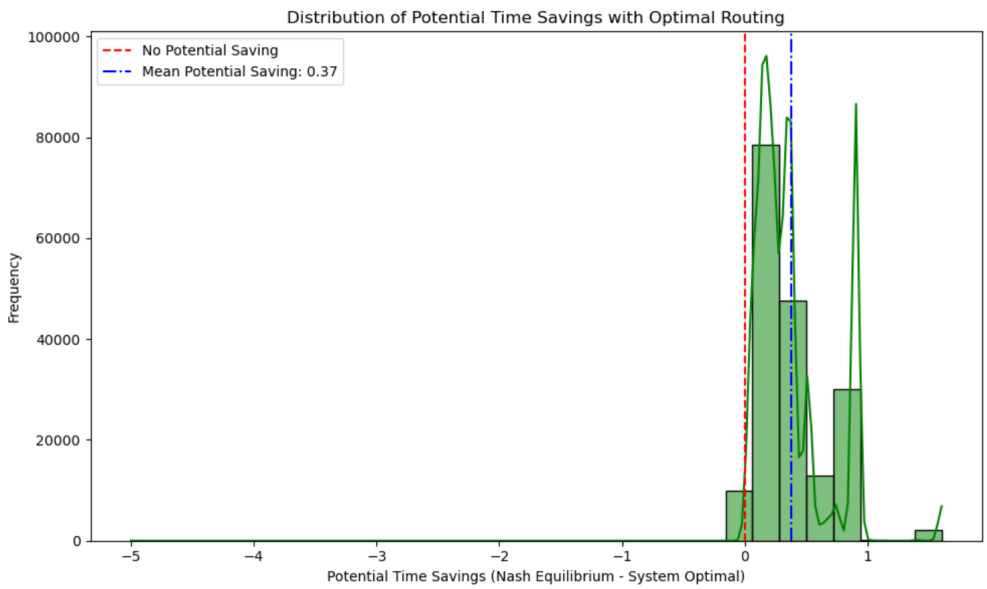


Nash Equilibrium Metrics

Our Nash equilibrium simulation yielded more severe congestion results than the observed traffic conditions. While the average Price of Anarchy (PoA) remains at 10.05, the Nash equilibrium delay increased to 69,927.76 units (compared to the observed 51,062.65 units), representing a 37% higher total delay. The average Nash equilibrium delay per route grew to 0.39 units (compared to 0.28 units observed).

This significant difference suggests that if all drivers were to follow purely selfish routing strategies with perfect information, London's congestion would be even worse than current conditions. This finding indicates that some degree of sub-optimal route choice (due to incomplete information or other factors) may actually be preventing the worst-case Nash equilibrium scenario from manifesting.





Potential Improvements Through Optimal Routing

Our analysis indicates significant potential for improvement through system-optimal routing. On average, each route could save approximately 0.37 time units if optimal routing were implemented instead of the Nash equilibrium patterns.

However, these findings must be interpreted with several important caveats:

1. It assumes perfect compliance from all drivers, which is unrealistic without strong incentives or enforcement.

2. It requires centralized route assignment, which raises practical and privacy concerns.

3. It does not account for the dynamic nature of traffic, where new drivers continually enter the system.

The contrast between Nash equilibrium and system-optimal travel times underscores the fundamental challenge of traffic management. When drivers make individually rational decisions to minimize their own travel times, they collectively create an inefficient equilibrium that significantly increases travel times for everyone. This "tragedy of the commons" in urban mobility explains why initiatives like congestion charging and ULEZ have been implemented in London.

Limitations and Caveats

1. **Representation Issues**: Our sample of 975 OD pairs captures only a fraction of all possible trips in London. The selected OD pairs may not fully represent all travel patterns, particularly for local or infrequent journeys.

2. **Simplified Cost Functions**: Our models used approximations of selfish and optimal cost functions that may not fully capture the complexity of real-world traffic dynamics. The marginal cost factor in the optimal cost function (0.10) is not empirically calibrated, potentially affecting the accuracy of potential time savings estimates.

3. **Static Analysis**: Our approach uses a static traffic assignment model that does not account for the temporal dynamics of traffic flow. In reality, traffic patterns evolve throughout the day, and drivers adapt their routes based on real-time conditions.

4. **Perfect Information Assumption**: The Nash equilibrium model assumes that drivers have perfect information about travel times on all possible routes, which is rarely the case in practice. Information asymmetry can lead to patterns that deviate from theoretical equilibria.

5. **Implementation Challenges**: While our analysis suggests huge potential benefits from system-optimal routing, achieving this in practice would require overcoming significant behavioral, technological, and political challenges.

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